INCH-POUND
MIL-DTL-24643/25G
27 November 2023
SUPERSEDING
MIL-DTL-24643/25F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +302 DEGREES FAHRENHEIT (+150 DEGREES CELSIUS), 300 VOLTS, TYPE LSPI

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Non-Watertight with Circuit Integrity)

First	-	Copper conductor, nickel-coated, AWG 18, class B stranding.
Second	-	Extruded silicone rubber insulation. Minimum average wall thickness of 0.018 inch.
Third	-	Glass braid.
Fourth	-	Braid covering.
Fifth	-	Two conductors cabled together to form a pair with a lay not greater than 2.25 inches. Conductors color coded by standard identification code method 4 (one black and one white for each pair of conductors) with transparent braid covering, or by white (natural) glass braid covered with colored braid covering (one black and one white for each pair of conductors).
Sixth	-	An optional binder.
Seventh	-	Braided shield of 34 or 36 AWG uncoated copper over each pair. Minimum coverage of 85 percent.
Eighth	-	Shield insulation over each pair shield, consisting of polyethylene terephthalate film, 0.001 inch thick, applied helically with 50 percent overlap.
Ninth	-	A glass braid over each shield insulation (standard identification code applied by method 4).
Tenth	-	The specified number of pairs (see table I) cabled together with a lay not greater than 14 inches. Fillers may be used as necessary to form a firm, well-rounded assembly. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward.
Eleventh	-	A binder applied helically with overlap.
Twelfth	-	Silicone rubber jacket colored orange or red (see table I for thickness). Jacket surface markings not applicable.
Thirteenth	-	Braided metal armor.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/25	Type and size	No. of pairs	Overall cable (inches)	
			Jacket thickness (min. avg.)	Diameter (max.)
-01AN	LSPI-3	3	0.050	0.685
-02AN	LSPI-7	7	0.063	0.900
-03AN	LSPI-12	12	0.063	1.155

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F (25 °C), max.)	7.5
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	1,200
Conductor to shield	1,200
Shield to shield	500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Shield to shield	100
Conductor continuity	No failure
Shield continuity	No failure

Group A:

Visual and dimensional	No failure

NOTE: The nickel coating on the conductors of the completed cable shall be permitted to have minor nicks, scrapes, or abrasions that are just perceptible to the eye.

Abrasion resistance (insulation, scrapes min.)	250
Crack resistance	No damage

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50

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Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	800
Elongation (percent, min.)	250
Durometer (jacket) – type A (hardness, min.)	60

Group C:

Physicals (cable aging) (500±18 °F [260±10 °C])	
Jacket (cable)	
Tensile strength (lb/in^2, min.)	600
Elongation (percent, min.)	150
Shield (conformance to material, construction, and coverage)	No failure
Armor (conformance to material, construction, and coverage)	No failure

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Gas flame (circuit integrity) (3 hours)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Armor	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2

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Fillers	0.2
Insulation	0.2
Smoke index (max.)	
Jacket	35
Fillers	45
Insulation	35
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (min.)
LSPI (all sizes)	100

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING ACTIVITY

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-041)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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